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Deficiency and toxicity of boron: Alterations in growth, oxidative damage and uptake by citrange orange plants

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ABSTRACT

Boron (B) deficiency and toxicity are the major factors that affect plant growth and yield. The present study revealed the effect of B deficiency and toxicity on plant growth, morphology, physiology, and cell structure. A hydroponic culture experiment was conducted with five B levels, B deficient (B0), sufficient (B20, B10, B40) and toxic (B100). Our results show that both B deficient as well as excess level inhibit plant growth. In B deficiency, the major visible symptoms were appeared in roots, while B excess burned the leaf margin of older leaves. The antioxidant enzymes including superoxide dismutase (SOD), peroxidase (POD), catalase (CAT) and ascorbate peroxidase (APX) decreased at B deficiency and also decreased up to some extent at B excess, while in sufficient treatments, the higher antioxidant enzymes were found at B20. In addition, the MDA concentration decreased at B deficiency and increased with B concentration. Moreover, the photosynthetic rate, transpiration rate, stomatal conductance, leaf gas exchange and intercellular CO_2 were reduced at both B deficiency as well as excess and higher at sufficient B20 treatment significantly. The chlorophyll and carotenoid content increased at B20 treatment, while decreased at B deficiency and excess. The middle lamellae of cell wall were found thick at B excess and normal at B20. The current study revealed that B deficiency as well as excess concentration affect plant growth and various morpho-physiological processes.

1. Introduction

Boron (B) is an important micronutrient necessary for the plant growth and development (Tanaka and Fujiwara, 2008). Although the requirement of B varies in different crops but the optimum value of B content for most crops is about 20 mg/kg^{-1} The availability of B concentration in proper amount in soil is necessary for plant growth, development and yield. In the earth crust B content availability ranges from 20 to 200 μg g^{-1}, and B available in soils ranges between 0.4 and 5 μg g^{-1}. The average B content in plant tissues is to be found about 2 μmol g^{-1} (20 μg g^{-1}) of dry weight. Plants uptake B as an un-dissociated boric acid (Hänsch and Mendel, 2009; Mengel and Kirkby, 2001; Tewari et al., 2010). The range of B between deficiency and excess, both is very narrow producing toxic symptoms that affect crop yield and production (Camacho-Cristóbal et al., 2008). B deficiency is a nutritional disorder which vary from high B soils and high rainfall areas all over the world (Marschner, 2012). Plants which grow in B deficient soil must show B deficiency symptoms in initial stage, such as abnormal growth of the apical growing region. Later on, plants show retarded growth, because B deficiency affect physiological, biochemical and molecular processes and resulted in decreased yield (Tewari et al., 2010). Root growth is also sensitive to B deficiency, primarily as a result of disturbing cell wall plasticity and additionally due to the inhibition of root meristem functioning (Herrera-Rodríguez et al., 2010).

In addition to other plants, Boron deficiency and excess also affect citrus growth and productivity. For example, in southern part of China B deficiency negatively affect citrus quality and productivity as well (Jiang et al., 2009; Wang et al., 2011). B deficiency can be removed through the B application by various ways. Many previous studies showed that B application improved yield and fruit quality (Boaretto et al., 2011). In addition to B fertilizers, B deficiency can also be controlled by using different rootstocks to absorb B efficiently from the soil by citrus (Boaretto et al., 2008). On the other side, B excess also results in plant toxicity and also damage the plant growth and productivity. The primary symptoms exhibit by plants exposed to B toxicity are burning of older leaves margins or tips i.e., chlorosis and necrosis and

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reduce vigor. Later on, it delay development, inhibit plant growth and decrease weight, number and size of fruits (Herrera-Rodríguez et al., 2010; Reid and Fitzpatrick, 2009). It is well known from the previous studies that B toxicity increases oxidative stress (Aftab et al., 2010; Cervilla et al., 2007; Landi et al., 2014). For instance, it has been described that excess of B from contaminated soil causes oxidative stress in sunflower (*Helianthus annuus* L.) (Giansoldati et al., 2012; Tassi et al., 2011).

Citrus belong to the genus *Citrus L.* of the family Rutaceae. It is subtropical evergreen fruit tree that are mostly cultivated on acid soils in humid and sub humid of tropical, subtropical, and temperate areas of the world. Citrus count as one of the important commercial crops in China as well as all over the world. The annual production of citrus is about 102 million tons (Mehl et al., 2014). Citrus fruits are highly accepted and appreciated by consumers throughout the world because of their pleasant flavors, attractive color and aroma. With the increase in production, improvement in storage, processing techniques and year-round supply, citrus fruits became an important commercial fruit and dietary source of nutrients for people all over the world. Previous reports showed that B deficiency commonly observed in China, and is responsible for significant loss of citrus fruit productivity and quality (Han et al., 2009; Jiang et al., 2009; Xiao et al., 2007).

In order to adapt their self in harsh environmental conditions, citrus plants evolved various processes to cope with numerous stresses including B deficiency and excess. Antioxidant enzymes are one of these strategies that tolerate plant against multiple stresses including B deficiency and toxicity (Juan et al., 2005). As in most environmental stresses, B toxicity leads to the formation of reactive oxygen species (ROS) which cause oxidative damage of cellular membranes and finally the cell death may occur (Cervilla et al., 2007). In order to reduce cellular damage produced by ROS, plants have developed the scavenging systems composed of antioxidant enzymes, such as superoxide dismutase (SOD), peroxidase (POD), ascorbate peroxidase (APX) and catalase (CAT) (Chen et al., 2005), and non-enzymatic scavenging system like ascorbate, glutathione, alpha -tocopherol (Blokhina et al., 2003; Miller et al., 2008) and osmoprotective solutes like proline (Cervilla et al., 2012). It has been well reported that under B deficiency plants display antioxidant enzymes to scavenge ROS (Karabal et al., 2003; Gunes et al., 2006; Molassiotis et al., 2006; Han et al., 2008; Tewari et al., 2010).

The present work was conducted to study the effects of B toxicity and deficiency on the growth of citrange orange plant. In addition, B concentrations in leaf, stem and root were also determined. Moreover, lipid peroxidation (MDA), proline and H_2O_2 content, and antioxidant enzyme (SOD, POD, CAT and APX) activities were also checked to determine the optimum concentration required for citrange. This study was designed to provide a basis for developing strategies to reduce the risks related to B deficiency and toxicity and maintain and/or increase citrange production.

2. Material and methods

2.1. Plant materials, growth conditions and treatments

Hydroponic conditions were used for this study. Six month-old Citrange plants with uniform size were brought from a commercial plant market. To remove surface contaminants, seedlings were washed three times with distilled water and soaked in tap water for two days. The resultant seedlings were transplanted in black pots containing three liter of nutrient solution. Black foam board with holes for seedling were used to fix the seedlings and put in an illuminated greenhouse with relative humidity of 60–80%, a constant temperature of 26 °C, and a photoperiod of 14 h/10 h light/dark (300–320 μmol m^{-2} s^{-1} photosynthetic photon flux density). The greenhouse, where experimental plants were placed, is located in Huazhong Agricultural University, Wuhan, China. Before the experimental treatments, modified form of Hoagland and Arnon (0.5 mM KNO_3, 0.6 mM $Ca(NO_3)_2$, 0.5 mM $MgSO_4$, 0.07 mM Na_2HPO_4, 0.16 mM NaH_2PO_4, 2.2 μM $MnCl_2$, 0.4 μM $ZnSO_4$, 0.08 μM $CuSO_4$, 0.09 μM Na_2MoO_4, 14.4 μM EDTA-Fe) were provided for two weeks. Meanwhile, two-week-old seedlings grown in the same conditions were kept as control. The experiments were started on April 2016, and lasted 75 days.

2.2. Plant growth parameters

The primary root length was measured with the help of a ruler from the top of the hypocotyls along the taproot to the root tip. In addition, plant height was measured by a ruler from the bottom of the epicotyls along the stem to shoot tips.

2.3. Leaf gas exchange

The leaf gas exchange parameters such as the photosynthetic rate (Pn; lmol CO_2 m^{-2} s^{-1}), intercellular CO_2 concentration (Ci; mmol mol^{-1}), stomatal conductance (gs; mmol H_2O m^{-2} s^{-1}), and transpiration rate (E; mmol H_2O m^{-2} s^{-11}) were measured by using the Li 6400 portable photosynthesis system (LI-COR Inc., Lincoln, NE, USA) at the end of the experiments. Measurements were conducted on the basal leaves of plant which are located between the middle and the base of the scion's shoots. These parameters were measured from 09:00 a.m. to 12:00 p.m. under balanced light intensity of 1000 lmol m^{-2} s^{-1} with 385 ± 10 llL^{-1} CO_2 concentration and 28–30 °C temperature.

2.4. Plant morphology and B uptake by citrange

The root scanning analyzer (WinRHIZO2009) was used to observe the root morphology. All the plant parts i.e., root, stem and leaves were washed consecutively with tap water and deionized water for at least three times. The corresponding samples were dried in an oven at 70 °C for 3 days. After recorded fresh and dry weight of samples, the subsequent dry samples were then grounded to fine powder and stored for further analyses.

To determine the B concentration in root, stem and leaves, 0.2 g grounded powder of each part were ashed at 550 °C in porcelain crucibles and followed by dissolution of 10 ml 0.1 M HCl. Sheng et al. (2009) was further followed to determine the B concentration in citrange.

2.5. Antioxidant enzyme activity

To evaluate the enzymes activities in shoots of different B treatments, freshly washed above ground part (0.5 g) were ground in liquid N_2 and homogenized in 5 ml of 0.2 mol L^{-1} sodium phosphate buffer (pH 7.8). The homogenate was then centrifuged at 12,000 rpm for 15 min at 4 °C, and the supernatant was collected for the determination of enzymes activities.

SOD activity was evaluated by the method of Beauchamp and Fridovich (1971). CAT activity was assessed by using the method of Aebi and Bergmeyer (1983). POD activity was checked by using the guaiacol oxidation method as illustrated by Li (2000). APX activity was evaluated by the method of Nakano and Asada (1981). Lipid peroxidation was measured by determination of the MDA contents as described by Heath and Packer (1968).

2.6. Microscopy

For (Transmission Electron Microscope) TEM study, the samples were fixed in 2.5% glutaraldehyde solution in 50 mM potassium phosphate buffer (pH 6.8) and kept overnight at room temperature. After that, samples were washed three times, for 15 min each, with 50 mM sodium cacodylate buffer (pH 6.9) and ultimately diluted (1:1) with 50 mM sodium cacodylate buffer (pH 6.9) (Sandalio et al., 2001).

The leaf chlorophyll and carotenoid contents were measured by using the method Mackinney (1941).

2.7. Statistical analysis

These experiments were carried out in complete randomize design with four replicates and each replicate represents at least nine plants. The mean of independent plants per treatment was used for standard deviation of growth parameters, root morphology, antioxidant enzymes activity, and MDA content leaf gas exchange and other parameters determined in these two experiments. Results were submitted to an analysis of variance (ANOVA) using STATIX 8.1. When the ANOVA showed a statistical effect, means were separated by least significant differences (LSD) at $P < 0.05$.

3. Results

3.1. Plant growth and symptoms under B deficiency and toxicity

Our results showed that the plant heights after 30 days under B deficient and excess levels were 8.6 and 9.8 cm respectively, which were considerably lower than 11.7 cm as control (B20) treatment. After 57 days of treatment, plant growth was completely stopped under B deficient (B0) while, plants under B excess (B100) continue their growth throughout the trial. Plant height for B0, B10, B20, B40 and B100 were 18.6, 22.3, 24, 19.8 and 18 cm respectively at the end of the experiment (Table 1). It reveals that both B deficiency and excess treatments suppressed the plant growth at early stage of B treatment, however only B deficient treatment affect plant growth at later stages of the experiment. In contrast, the plant height and growth was good at normal (B20) treatment throughout the experiment.

The results obtain in this study revealed that in addition to plant height; low and high B concentration in culture solution also alter the number and dry weight of leaves, root, stem and finally the whole citrange plant (Tables 2 and 3). Dry weight of root, stem and leaves under B deficient concentration were 0.38, 0.41 and 0.23 g respectively, whereas the dry mass of these parts under B excess treatments were 0.41, 0.44 and 0.25 g. All these value were found significantly lesser than B20 treatments which were 0.63, 0.59 and 0.35 g for root, stem and leaves respectively. It indicates that both B deficiency and toxicity significantly suppressed the citrange growth parameters like root, stem and leaves (Tables 1–3). Moreover, our results revealed that B deficiency as well as excess limited the growth of root system mainly. As B concentration were increased from deficiency to normal, the plant height was increased, number of leaves was observed higher and the plant biomass were also increased. But when the concentration increased from normal to excess, the plant height again decreased and less number of leaves and total biomass were also decreased.

With the inhibition of plant growth, chlorosis appeared in older leaves under B excess treatment. Chlorosis started at the tips of leaves, then along the margins, and finally end with scorched leaves. These symptoms were not observed clearly but a little. For other treatments, these symptom were not observed.

Table 1
Plant height (cm), primary root length (cm), leaf area (cm) and number of leaf under different B levels.

B Level	Plant height	No of leaves	Primary root length	Leaf area
B0	18.63 ± .64^{b}	17.67 ± .206^{b}	8.57 ± .941^{b}	4.84 ± 1.3^{b}
B10	22.33 ± .28^{ab}	22 ± .032^{a}	10.33 ± 2.21^{b}	5.72 ± 1.1^{ab}
B20	24 ± .301^{a}	24.33 ± .551^{a}	13.23 ± 2.51^{a}	7.11 ± 1.2^{a}
B40	19.87 ± .071^{b}	19.5 ± .411^{b}	11.57 ± 1.9^{ab}	4.84 ± .85^{b}
B100	18.81 ± .05^{b}	18.33 ± .610^{b}	9.93 ± 2.4^{b}	4.46 ± .52^{b}

Different letters behind S.E indicate significant difference at P ≤ 0.05 by LSD (n = 3).

3.2. Boron uptake

During all treatments, B uptake in different parts of plants showed a common pattern of new leaves > root > stem (Fig. 1). Lowest B concentration was found in the stem, while highest concentration was found in the new leaves during all treatments. When the concentration of B increased, its uptake also increased in all parts of plant. The highest B concentration was found 21.87 mg/kg in leaves during B excess. On the other hand, the lowest concentration of B was 4.9 mg/kg under B deficient treatment, which is still very low due to some environmental stresses. Although, B concentrations were low in root and stem but its concentration increased with the increased B supply, while degree of increase still remained at a relatively low level.

3.3. Root morphology

Our results revealed that when the B concentration increased from deficient to normal level, total root length, volume, surface area and root forks also enhanced. In contrast, when the B treatment were further increased from normal to excess level, the root parameters were decreased. The average root diameter decreased, when the concentration of B increased upto B20, but in B40 and B100 treatments it start increasing again. A significant decline was observed in root tips under B deficiency treatment and a significant increased in root tips under B excess (Table 3).

3.4. Antioxidant enzyme activity

A positive relationship among the boron concentration and antioxidant enzymes activities were found in all treatments of B to citrange (Fig. 2). The SOD activity decreased under B deficiency (B0) and excess (B100), while high at B20 treatment significantly. SOD results also normal at B10 and B40 treatments. The highest value in SOD is 206.6 (ug^{-1} FW) at B20 treatment and the lower value is at B deficient treatment (B0) 59.5 (μg^{-1} Fw) (Fig. 2a). The activity of CAT, POD and APX showed the same manner (Fig. 2b, c, d). The CAT and POD activities decreased significantly at both B deficiency and excess. Its values increased at B20 than other treatments. In other words, the antioxidant enzymes (SOD, POD, CAT and APX) increased from B0 to B20 and then it decreaced at treatment B100. Highest value for POD, CAT and APX were 2296 u/g min, 23.4 u/g FW and 2.55 mmol g^{-1}. In addition, our result showed that MDA concentration was lower at B deficient treatment and high at excess. At B deficiency, the MDA value is 2.36 umol/L, while at B excess its value is 6.10 umol/L (Fig. 2e).

3.5. Leaf gas exchange and chlorophyll and carotenoids contents

Boron deficiency and toxicity also influenced numerous leaf gas exchange parameters. At B deficient and excess treatments, stomatal conductance (gs), leaf photosynthetic rate (Pn), transpiration rate (E) and intercellular CO_2 concentration (Ci) was significantly lower than the plants grown at B sufficient level (Fig. 3a–d). Moreover, we observed relative higher values in Pn, E, Ci and gs in leaves of citrange at B sufficient treatment.

We observed that B deficiency and excess caused harmful affect to chlorophyll pigments. B deficiency significantly reduced chlorophyll (a, and b) and carotenoids contents (Fig. 4). On the other hand, B excess also influenced chlorophyll (a and b) and carotenoids contents in the citrange plants. At B20 treatment, chlorophyll (a and b) and carotenoids contents were recorded significantly high, while in other treatments such as, B10 and B40 were also showed good performance instead of B deficient and excess level (Fig. 4a–c).

3.6. Cellular study

Plant tissues were observed under TEM, palisade cells of B excess

Table 2
Fresh and dry weight (grams) of citrange orange plants under different boron treatments. 0 μM as a deficient, 100 μM as excess and 10 μM, 20 μM, and 40 μM as a normal treatments were used.

	Plant fresh weight			Plant dry weight			
Treatment	**Root**	**Stem**	**Leaf**	**Root**	**Stem**	**Leaf**	**Shoot/root**
B0	0.97 ± 0.095^{c}	0.83 ± 0.11^{b}	0.63 ± 0.045^{c}	0.38 ± 0.021^{c}	0.417 ± 0.034^{b}	0.23 ± 0.030^{c}	1.08 ± 0.21^{ab}
B10	1.23 ± 0.023^{ab}	1.09 ± 0.10^{ab}	1.00 ± 0.065^{a}	0.49 ± 0.041^{b}	0.52 ± 0.021^{ab}	0.32 ± 0.025^{a}	0.98 ± 0.07^{ab}
B20	1.47 ± 0.017^{a}	1.29 ± 0.07^{a}	1.21 ± 0.18^{a}	0.59 ± 0.011^{a}	0.63 ± 0.029^{a}	0.35 ± 0.029^{a}	0.96 ± 0.11^{ab}
B40	1.17 ± 0.11^{bc}	1.03 ± 0.04^{ab}	0.80 ± 0.071^{b}	0.49 ± 0.037^{b}	0.44 ± 0.012^{b}	0.27 ± 0.022^{b}	0.92 ± 0.097^{b}
B100	1.05 ± 0.064^{bc}	0.93 ± 0.02^{b}	0.73 ± 0.011^{b}	0.41 ± 0.032^{bc}	0.39 ± 0.020^{b}	0.25 ± 0.013^{bc}	1.15 ± 0.20^{a}

Values are the mean of four replicates ± SD. Different letters in each rows indicate significant difference ($P < 0.05$).

leaves were conspicuous by the presence of thick cell wall middle lamellae, but such features were not found in the palisade cells of B sufficient leaves and B deficient the middle lamellae were not observed. The nucleus, chloroplast and other important part of plant cell in the leaf of citrange was not observed clearly (Fig. 5a). Furthermore, in B20 treatment the middle of cell wall is normal show on the red arrow and the nucleus and chloroplast is also normal showing with the help of green arrow (Fig. 5b). In addition, the (Fig. 5c) the cell wall middle lamelle is thick as compared to B20 treatment and other part of leaf cell is normal

4. Discussion

It is well known that B is an important element for plant growth and development, and the primary response to B deficiency is the inhibition of root elongation which influence the main and lateral roots (Dell et al., 1997). In addition, B deficiency and toxicity cause numerous other morphological changes including the browning of leaves, death of apical meristems, abnormal enlargement of root tips, dense appearance of lateral roots near to the root apex, lack of root hairs and growth inhibition (Goldbach et al., 2001). Moreover, it also affects physiological, biochemical and molecular processes. In our case study, taproot growth of citrange orange was inhibited in early days of B deficiency, and then completely stopped after 57 days. It represents that in early stage of the experiment, B was adequate to keep plant healthy at the early stages of their growth. However, B is regularly in demand for the growth and development, while insufficient supply of B appear B deficiency symptoms in citrange plants (Uraguchi et al., 2011, Alves et al., 2011). In addition, B deficiency also reduced total root length, the number of root tips, total root volume and total root surface area of citrange seedlings (Table 2). Furthermore, number of leaves, dry weight of leaves and stem was also decreased with B deficiency (Tables 2 and 3). B deficiency inhibits the root growth, formation of the new roots, number of leaves, dry weight of stem and leaves, and finally the whole citrange plant. Similar B deficiency symptoms were also observed in tomato and Trifoliate oranges seedling including the root color became darker, root apical enlargement and lateral root number increased near the apical part (Wang et al., 2013)

On the other hand, B excess also affects plant growth and development. Leaf chlorosis was observed first at the leaf tips, then expands

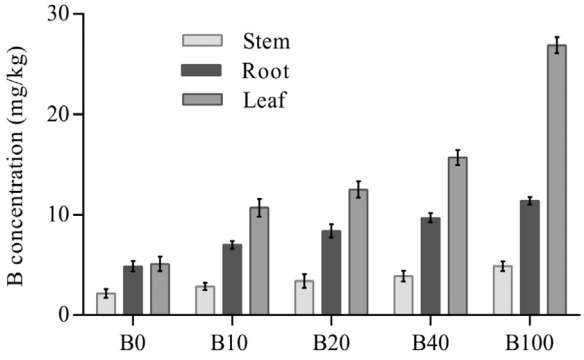


Fig. 1. Boron (milligram per liter dry weight) uptake by citrange orange plant. Leaves uptake the highest amount of B under all treatments. 0 μM as a B deficient level, 100 μM as an excess level and 10 μM, 20 μM, 40 μM as a normal treatment. *Error bars* represent ± SD of four biological replicates.

to the margins and finally whole leaves. As compared to younger leaves, old leaves were affected earlier by B toxicity. Han et al. (2009) and Sheng et al. (2010) also found similar symptoms in barley and citrus respectively. B toxicity symptoms do not appear in roots visibly which suggests that B distribution is related to the transpiration stream. B mainly accumulated in citrange leaves, but in relatively low concentrations it also remains in the roots as well. Although, there were no visible symptoms in roots under B excess, but overall its length, surface area and volume decreased significantly (Table 2). Various reports are in line with our study that the root length and diameter decreased under B excess which indicates that B toxicity significantly inhibit root growth (Choi et al., 2007; Sheng et al., 2010; Reid, 2007). Moreover, B excess also significantly suppressed the plant height. We suggests that overall B excess significantly inhibit the citrange growth. The most possible mechanism for reduction in root can be interference with mitosis (Reid, 2007). More considerably, oxidative damage by hydrogen peroxide and lipid peroxidation may have negatively affected the photosynthetic system and consequently affect plant growth including plant height, root volume and biomass. We also observed decreased transpiration rate, stomatal conductance, photosynthetic rate and intercellular CO_2 at both B deficiency and excess in citrange plant (Fig. 3). Similar reports were found in other citrus plants (Papadakis et al.,

Table 3
Effects of different B treatments on citrange orange plant. Root length (cm), average root diameter (mm), total root volume (cm^3), total root surface area (cm^2), root tips and root forks of citrange orange were measured under different levels of B concentrations.

B Level	**Total root length**	**Total root volume**	**Root tips**	**Total surface area**	**Root forks**	**Average root diameter**
B0	134.48 ± 20.2^{d}	0.819 ± 0.120^{c}	51.83 ± 9.21^{c}	13.33 ± 1.4^{b}	161.2 ± 22.2^{c}	1.37 ± 0.20^{a}
B10	290.07 ± 15.6^{ab}	2.043 ± 0.082^{a}	122.67 ± 3.6^{b}	20.87 ± 1.6^{a}	500.6 ± 40.2^{b}	1.26 ± 0.051^{bc}
B20	318.61 ± 10.3^{a}	2.12 ± 0.101^{a}	137.5 ± 4.52^{a}	21.47 ± 1.0^{a}	624.67 ± 29.3^{a}	0.965 ± 0.062^{c}
B40	263.32 ± 17.45^{b}	1.78 ± 0.13^{a}	139.83 ± 13.2^{a}	20.55 ± 1.2^{a}	580.6 ± 48.8^{a}	1.07 ± 0.088^{bc}
B100	199.63 ± 21.8^{c}	1.39 ± 0.23^{b}	149.33 ± 16.1^{a}	19.93 ± 2.3^{a}	460.06 ± 34.5^{c}	1.16 ± 0.02^{b}

Different letters indicate significant difference at $P < 0.05$ by LSD test. (Mean ± SD).

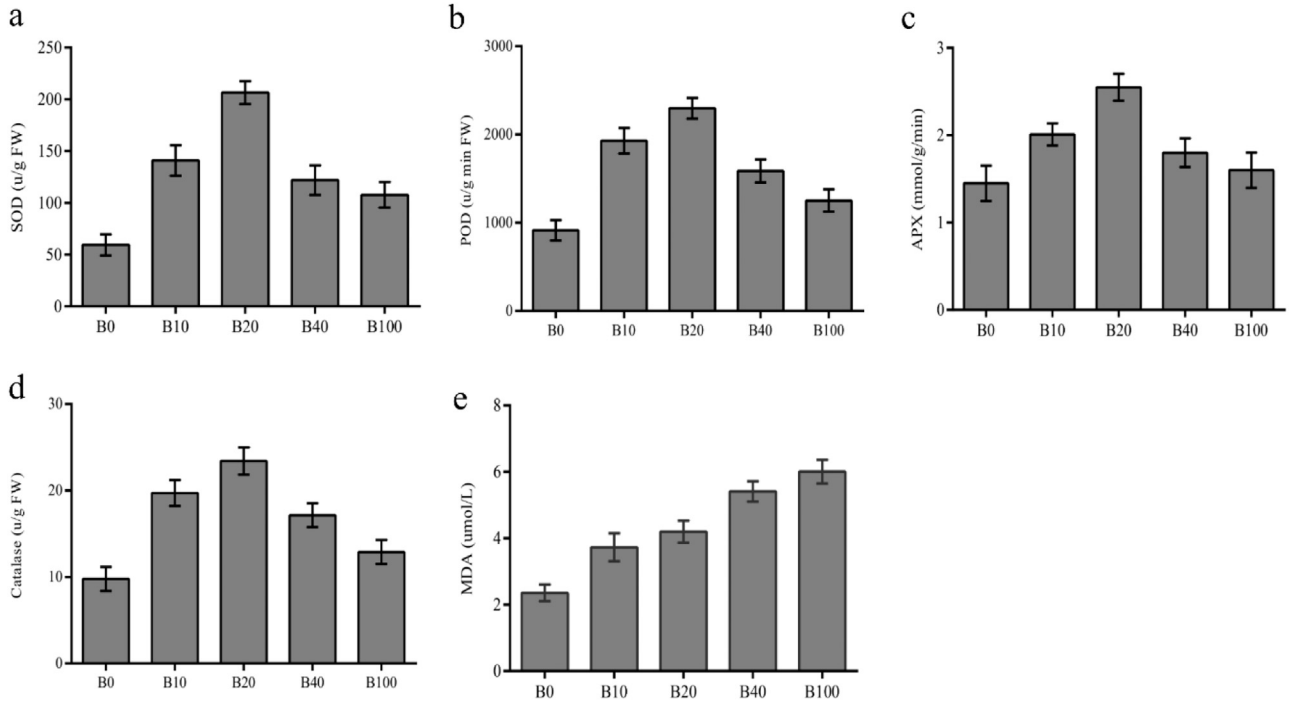


Fig. 2. Effect of B treatments on antioxidant enzyme activities. (a, b, c, d) SOD, POD APX and CAT were high at B20 and lower under B deficient and excess in leaves. (e) MDA content increased with the increase of B concentration. *Error bars* represent ± SD of four biological replicates. 0 μM as a deficient B level, 100 μM as excess B level and 10 μM, 20 μM, 40 μM as a normal B level.

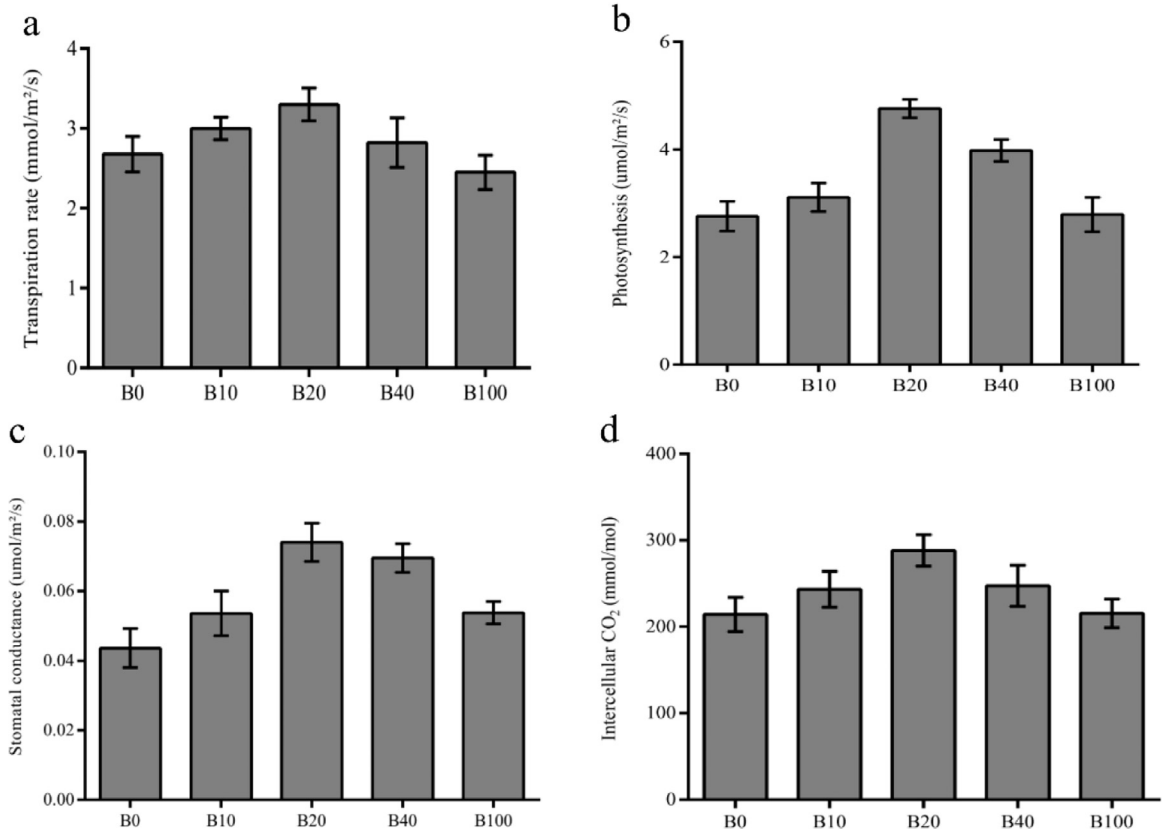


Fig. 3. Effect of boron on transpiration rate photosynthesis, stomatal conductance and intercellular CO_2 under different B levels. (a, b, c, d) Transpiration rate, photosynthesis, stomatal conductance and intercellular CO_2 were high under B20 i.e., B sufficient treatment, while under B deficient and excess level these were lower. *Error bars* represent ± SD of four biological replicates. 0 μM as a deficient B level, 100 μM as excess B level and 10 μM, 20 μM, 40 μM as a normal B level.

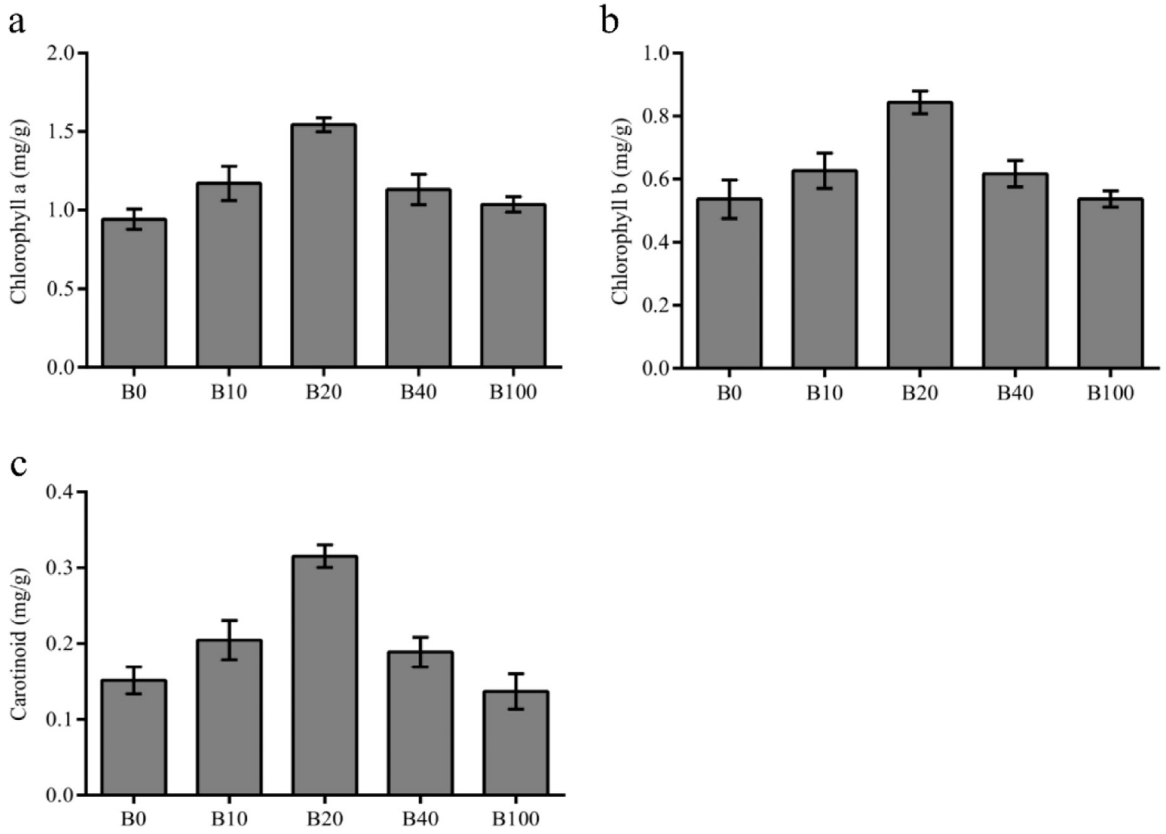


Fig. 4. Effect of B on chlorophyll a, chlorophyll b and carotenoid content under different boron treatments. (a, b, c) chlorophyll a, chlorophyll b and carotenoid content were high under normal concentration of B i.e., B20, while under deficient and excess level these parameters were low. *Error bars* represent ± SD of four biological replicates. 0 μM as a deficient B level, 100 μM as excess B level and 10 μM, 20 μM, 40 μM as a normal B level.

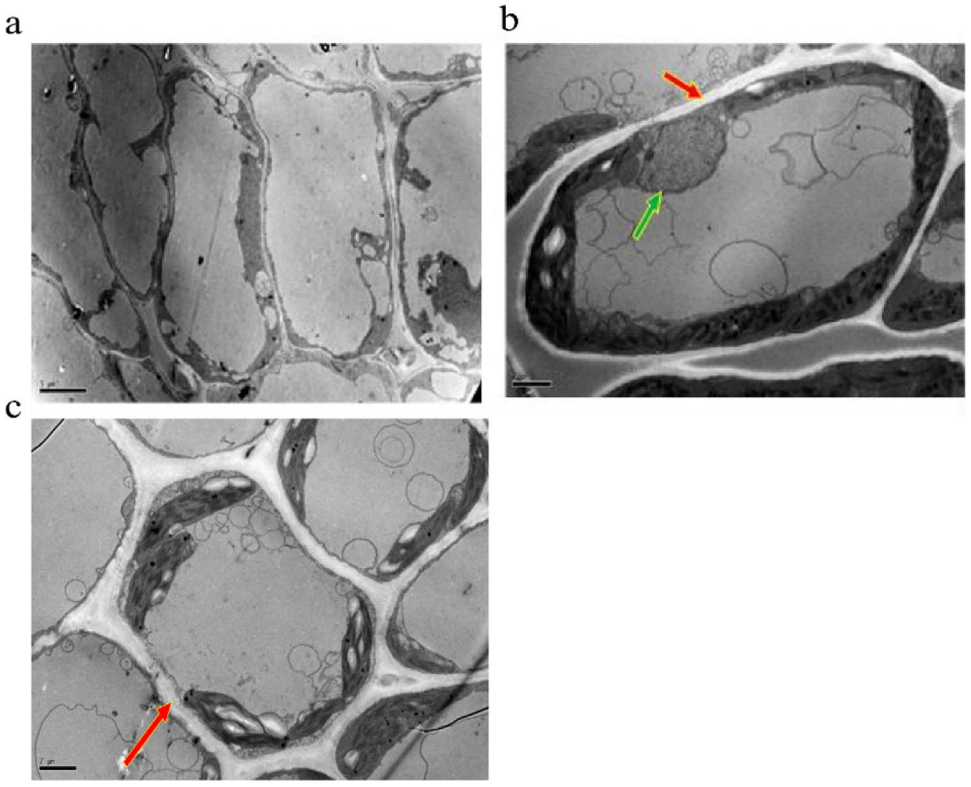


Fig. 5. Ultra structural changes in citrange leaf under B deficiency, excess and sufficient. (a) Cell walls of palisade cells under B deficient (0 μM) were not clearly observed. (b) Cell walls with normal middle lamellae and chloroplast under B 20 μM. (C) Excessive cell wall thickening in middle lamellae of palisade cells under B 100 μM treatment.

2004a, 2004b).

Antioxidant enzymes play important protective role in plants against various environmental stresses that cause ROS and oxidation of phenolic compounds (Mohamed and Shaaban, 2004; Fahad and Bano, 2012; Fahad et al., 2013, 2014a, 2014b, 2017; Saud et al., 2017). In this regard, various antioxidant enzymes activities were checked to identify the possible mechanism under which B inhibit growth and development. ROS are produced under stress; they are harmful, unless effectively scavenged by antioxidant enzymes. So, the activity of the enzymes associated with the antioxidant system may be considered as stress indicators in plants, due to their up regulated activities merely in the B stressed citrange plants. Effects of B deficiency on the activities of antioxidant enzymes has been well documented (Han et al., 2008). Our results showed that CAT activity increased at B deficient treatment and decreased at B excess. The CAT activity was also found higher in mung bean plants treated with Pb and Cd (Hassan and Mansoor, 2014). Moreover, we also observed decreased antioxidant enzymes such as SOD, POD and APX activity under B excess condition. It is revealed that antioxidant enzymes plays vital role in citrange tolerance to B deficiency and excess as plant growth parameters were significantly affected by B excess condition. Moreover, the increase of antioxidant enzymes significantly decreased MDA content. The MDA reduction at B deficiency seems to be associated with the induction of APX activity and improvement of other antioxidant enzymes activities. The MDA level and H_2O_2 concentration was higher in excess condition. With the increase in B concentration, the level of MDA became higher.

5. Conclusion

Collectively we concluded that both B deficiency and toxicity alter morphology, physiology, biochemistry and cellular structure of citrange plants. It cause inhibition of root growth, plant height, plant biomass and finally decrease plant growth, At B deficiency and excess the plant antioxidant enzyme system also affected. The SOD, POD, CAT and APX activities are lower at both B excess and deficient treatment. The MDA content damage the plant under B deficiency. The leaf gas exchange also influenced during B deficiency and excess conditions. Furthermore, the chlorophyll and carotenoid content also decreased under B deficiency and excess. The antioxidant and leaf gas exchange are normal at B20 treatment and the plant grow well and healthy at this level.

Author contributions

C.C.J. designed and supervised this study. A.S. conducted the experiments, performed data interpretation and drafted the manuscript. M.R helped in the experiments. W.X.W. and L.Y. helped in interpreting the results of the study, and L.Y. helped replaces nutrition solution in the experiment and determines B concentration. A.U and S.F revised the manuscript critically. All authors read and approved the final manuscript.

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References

Aebi, H.E., Bergmeyer, H.O., 1983. Catalase, Methods Enzymology. Academic, New York.
Aftab, T., Khan, M., Idrees, M., Naeem, M., Ram, M., 2010. Boron induced oxidative stress, antioxidant defence response and changes in artemisinin content in *Artemisia annua* L. J. Agron. Crop Sci. 196, 423–430.
Alves, M., Moes, S., Jenö, P., Pinheiro, C., Passarinho, J., Ricardo, C., 2011. The analysis of Lupinus albus root proteome revealed cytoskeleton altered features due to long-term boron deficiency. J. Proteom. 74, 1351–1363.
Beauchamp, C., Fridovich, I., 1971. Superoxide dismutase: improved assays and an assay applicable to acrylamide gels. Anal. Biochem. 44, 276–287.
Blokhina, O., Virolainen, E., Fagerstedt, K.V., 2003. Antioxidants, oxidative damage and oxygen deprivation stress: a review. Ann. Bot. 91, 179–194.
Boaretto, R.M., Quaggio, J.A., de Assis Alves Mourao Filho, F., Gine, M.F., Boaretto, A.E., 2008. Absorption and mobility of boron in young citrus plants. Commun. Soil Sci. Plant Anal. 39, 2501–2514.
Boaretto, R.M., Quaggio, J.A., Mattos Jr., D., Muraoka, T., Boaretto, A.E., 2011. Boron uptake and distribution in field grown citrus trees. J. Plant Nutr. 34, 839–849.
Camacho-Cristóbal, J.J., Rexach, J., González Fontes, A., 2008. Boron in plants: deficiency and toxicity. J. Integr. Plant Biol. 50, 1247–1255.
Cervilla, L.M., Blasco, B., Ríos, J.J., Romero, L., Ruiz, J.M., 2007. Oxidative stress and antioxidants in tomato (*Solanum lycopersicum*) plants subjected to boron toxicity. Ann. Bot. 100, 747–756.
Cervilla, L.M., Blasco, B., Rios, J.J., Rosales, M.A., Sánchez-Rodríguez, E., Rubio-Wilhelmi, M.M., Ruiz, J.M., 2012. Parameters symptomatic for boron toxicity in leaves of tomato plants. J. Bot. 2012.
Chen, L.-S., Qi, Y.-P., Liu, X.-H., 2005. Effects of aluminum on light energy utilization and photoprotective systems in citrus leaves. Ann. Bot. 96, 35–41.
Choi, E.Y., Kolesik, P., Mcneill, A., Collins, H., Zhang, Q., Huynh, B.L., Graham, R., Stangoulis, J., 2007. The mechanism of boron tolerance for maintenance of root growth in barley (*Hordeum vulgare* L.). Plant Cell Environ. 30, 984–993.
Dell, B., Huang, L., 1997. Physiological response of plants to low boron. Plant Soil 193, 103–120.
Fahad, S., Bajwa, A.A., Nazir, U., Anjum, S.A., Farooq, A., Zohaib, A., Sadia, S., Nasim, W., Adkins, S., Saud, S., Ihsan, M.Z., Alharby, H., Wu, C., Wang, D., Huang, J., 2017. Crop production under drought and heat stress: plant responses and management options. Front. Plant Sci. 8, 1147. http://dx.doi.org/10.3389/fpls.2017.01147.
Fahad, S., Bano, A., 2012. Effect of salicylic acid on physiological and biochemical characterization of maize grown in saline area. Pak. J. Bot. 44, 1433–1438.
Fahad, S., Chen, Y., Saud, S., Wang, K., Xiong, D., Chen, C., Wu, C., Shah, F., Nie, L., Huang, J., 2013. Ultraviolet radiation effect on photosynthetic pigments, biochemical attributes, antioxidant enzyme activity and hormonal contents of wheat. J. Food Agric. Environ. 11 (3–4), 1635–1641.
Fahad, S., Hussain, S., Bano, A., Saud, S., Hassan, S., Shan, D., Khan, F.A., Khan, F., Chen, Y., Wu, C., Tabassum, M.A., Chun, M.X., Afzal, M., Jan, A., Jan, M.T., Huang, J., 2014a. Potential role of phytohormones and plant growth-promoting rhizobacteria in abiotic stresses: consequences for changing environment. Environ. Sci. Pollut. Res. http://dx.doi.org/10.1007/ s11356-014-3754-2.
Fahad, S., Hussain, S., Matloob, A., Khan, F.A., Khaliq, A., Saud, S., Hassan, S., Shan, D., Khan, F., Ullah, N., Faiq, M., Khan, M.R., Tareen, A.K., Khan, A., Ullah, A., Ullah, N., Huang, J., 2014b. Phytohormones and plant responses to salinity stress: a review. Plant Growth Regul. http://dx.doi.org/10.1007/s10725-014-0013-y.
Goldbach, H.E., Yu, Q., Wingender, R., Schulz, M., Wimmer, M., Findeklee, P., Baluska, F., 2001. Rapid response reactions of roots to boron deprivation. J. Plant Nutr. Soil Sci. 164, 173–181.
Gunes, A., Soylemezoglu, G., Inal, A., Bagci, E., Coban, S., Sahin, O., 2006. Antioxidant and stomatal responses of grapevine (*Vitis vinifera L.*) to boron toxicity. J. Sci. Hortic. 110, 279–284.
Giansoldati, V., Tassi, E., Morelli, E., Gabellieri, E., Pedron, F., Barbafieri, M., 2012. Nitrogen fertilizer improves boron phytoextraction by Brassica juncea grown in contaminated sediments and alleviates plant stress. Chemosphere 87, 1119–1125.
Han, S., Chen, L.S., Jiang, H.X., Smith, B.R., Yang, L.T., Xie, C.Y., 2008. Boron deficiency decreases growth and photosynthesis, and increases starch and hexoses in leaves of citrus seedlings. J. Plant Physiol. 165, 1331–1341.
Han, S., Tang, N., Jiang, H.-X., Yang, L.-T., Li, Y., Chen, L.-S., 2009. CO_2 assimilation, photosystem II photochemistry, carbohydrate metabolism and antioxidant system of citrus leaves in response to boron stress. J. Plant Sci. 176, 143–153.
Hänsch, R., Mendel, R.R., 2009. Physiological functions of mineral micronutrients (Cu, Zn, Mn, Fe, Ni, Mo, B, Cl). Curr. Opin. Plant Biol. 12, 259–266.
Heath, R.L., Packer, L., 1968. Photoperoxidation in isolated chloroplasts: I. Kinetics and stoichiometry of fatty acid peroxidation. Arch. Biochem. Biophys. 125, 189–198.
Herrera-Rodríguez, M.B., González-Fontes, A., Rexach, J., Camacho-Cristóbal, J.J., Maldonado, J.M., Navarro-Gochicoa, M.T., 2010. Role of boron in vascular plants and response mechanisms to boron stresses. Plant Stress 4, 115–122.
Jiang, C., Wang, Y., Liu, G., Xia, Y., Peng, S., Zhong, B., Zeng, Q., 2009. Effect of boron on the leaves etiolation and fruit fallen of Newhall Navel Orange. Plant Nutr. Fertil. Sci. 15, 656–661.
Juan, M., Rivero, R.M., Romero, L., Ruiz, J.M., 2005. Evaluation of some nutritional and biochemical indicators in selecting salt-resistant tomato cultivars. Environ. Exp. Bot. 54, 193–201.
Karabal, E., Yücel, M., Öktem, H.A., 2003. Antioxidant responses of tolerant and sensitive barley cultivars to boron toxicity. Plant Sci. 164, 925–933.
Landi, M., Guidi, L., Pardossi, A., Tattini, M., Gould, K.S., 2014. Photoprotection by foliar anthocyanins mitigates effects of boron toxicity in sweet basil *(Ocimum basilicum)*. Planta 240, 941–953.
Li, H.S., 2000. Principles and Techniques of Plant Physiological Biochemical Experiment. Higher Education Press, Beijing (in Chinese).
Mackinney, G., 1941. Absorption of light by chlorophyll solutions. J. Biol. Chem. 140, 315–322.
Marschner, H., 2012. Marschner's Mineral Nutrition of Higher Plants. Academic Press, London.
Mehl, F., Marti, G., Boccard, J., Debrus, B., Merle, P., Delort, E., Baroux, L., Raymo, V.,

et al., 2014. Differentiation of lemon essential oil based on volatile and non-volatile fractions with various analytical techniques: a metabolomic approach. Food Chem. 143, 325–335.
Mengel, K., Kirkby, E.A., 2001. Principles of Plant Nutrition, 5th ed. International Potash Institute, Bern, Switzerland.
Miller, G., Shulaev, V., Mittler, R., 2008. Reactive oxygen signaling and abiotic stress. Physiol. Plant. 133, 481–489.
Mohamed, A., Shaaban, M., 2004. Nutrient status and enzyme activity alteration in cucumber seedlings as a response to boron deficiency. Acta Agron. Hung. 52, 9–17.
Molassiotis, A., Sotiropoulos, T., Tanou, G., Diamantidis, G., Therios, I., 2006. Boron-induced oxidative damage and antioxidant and nucleolytic responses in shoot tips culture of the apple rootstock EM 9 *(Malus domestica Borkh)*. Environ. Exp. Bot. 56, 54–62.
Nakano, Y., Asada, K., 1981. Hydrogen peroxide is scavenged by ascorbate specific peroxidase in spinach chloroplasts. Plant Cell Physiol. 22, 867–880.
Papadakis, I., Dimassi, K., Bosabalidis, A., Therios, I., Patakas, A., Giannakoula, A., 2004a. Effects of B excess on some physiological and anatomical parameters of 'Navelina'orange plants grafted on two rootstocks. J. Environ. Exp. Bot. 51, 247–257.
Papadakis, I.E., Dimassi, K.N., Bosabalidis, A.M., Therios, I.N., Patakas, A., Giannakoula, A., 2004b. Boron toxicity in 'Clementine'mandarin plants grafted on two rootstocks. Plant Sci. 166, 539–547.
Reid, R., 2007. Update on boron toxicity and tolerance in plants. In: Advances in Plant and Animal Boron Nutrition. Springer, Dordrecht, The Netherlands, pp. 83–90.
Reid, R., Fitzpatrick, K., 2009. Influence of leaf tolerance mechanisms and rain on boron toxicity in barley and wheat. Plant Physiol. 151, 413–420.
Saud, S., Fahad, S., Yajun, C., Ihsan, M.Z., Hammad, H.M., Nasim, W., Amanullah Jr., , Arif, M., Alharby, H., 2017. Effects of nitrogen supply on water stress and recovery mechanisms in kentucky bluegrass plants. Front. Plant Sci. 8, 983. http://dx.doi.org/10.3389/fpls.2017.00983.
Sheng, O., Song, S., Peng, S., Deng, S., 2009. The effects of low boron on growth, gas exchange, boron concentration and distribution of 'Newhall' navel orange (*Citrus sinensis* Osb.) plants grafted on two rootstocks. Sci. Hortic. 121, 278–283.
Sheng, O., Zhou, G., Wei, Q., Peng, S., Deng, X., 2010. Effects of excess boron on growth, gas exchange, and boron status of four orange scion-rootstock combinations. J. Plant Nutr. Soil Sci. 173, 469.
Sandalio, L.M., Dalurzo, H.C., Gomez, M., Romero Puertas, M.C., del Rio, L.A., 2001. Cadmium induced changes in the growth and oxidative metabolism of pea plants. J. Exp. Bot. 52, 2115–2126.
Tanaka, M., Fujiwara, T., 2008. Physiological roles and transport mechanisms of boron: perspectives from plants. Eur. J. Physiol. 456, 671–677.
Tewari, R.K., Kumar, P., Sharma, P.N., 2010. Morphology and oxidative physiology of boron-deficient mulberry plants. Tree Physiol. 30, 68.
Tassi, E.L., Pedron, F., Barbafieri, M., 2011. Evaluating the absorption of boron by plants—a potential tool to remediate contaminated sediments from Cecina River Basin in Italy. Water Air Soil Pollut. 216, 275–287.
Uraguchi, S., Fujiwara, T., 2011. Significant contribution of boron stored in seeds to initial growth of rice seedlings. Plant Soil 340, 435–442.
Wang, R.-D., Jiang, C.-C., Liu, G.-D., Wang, Y.-H., Peng, S.-A., Zeng, Q.-l., 2011. Status and analysis on available boron content in soil of gannan navel orange orchards. South China Fruits 4, 05.
Wang, R.-D., Liu, G.-D., Liu, L.-C., Jiang, C.-C., 2013. Effects of boron deficiency and excess on rootstock growth and root morphology in trifoliate orange seedlings. Int. J. Food Agric. Environ. 11, 817–823.
Xiao, J.X., Yan, X., Peng, S.A., Fang, Y.W., 2007. Seasonal changes of mineral nutrients in fruit and leaves of 'Newhall'and 'Skagg's Bonanza'navel oranges. J. Plant Nutr. 30, 671–690.